

LESSON 5-5 • SESSION 1 • PRACTICE • FORM A

Volume of Rectangular Prisms

Learning target – I can find the volume of a rectangular prism by multiplying length, width, and height. · 6.G.A.2

REMEMBER

- **Volume** is the space inside a solid, counted in unit cubes. Write **cubic units**: cm^3 , in^3 , ft^3 .
- One layer = length $\times$ width = **base area B**. The height counts the layers.
- **$V = l \times w \times h$ or $V = B \times h$.**

WORKED EXAMPLE

A box is 6 cm long, 3 cm wide, and 4 cm tall.

Step 1 One layer: $6 \times 3 = 18$ cubes.**Step 2** 4 layers stack up.**Step 3** $18 \times 4 = 72$.**Answer $V = 72 \text{ cm}^3$** **WORDS** **volume** (*volumen*) · **rectangular prism** (*prisma rectangular*) · **cubic units** (*unidades cúbicas*)**PRACTICE**

show your work · label units

1. Find the volume of each rectangular prism.

a. $8 \text{ cm} \times 6 \text{ cm} \times 9 \text{ cm}$ ____ cm^3

b. $7 \text{ ft} \times 4 \text{ ft} \times 3 \text{ ft}$ ____ ft^3

c. $9 \text{ ft} \times 6 \text{ ft} \times 2 \text{ ft}$ ____ ft^3

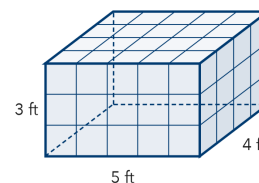
d. $B = 42 \text{ in}^2$, $h = 7 \text{ in}$ ____ in^3

- 2.
- Toy Chest.**
- A toy chest is packed with cubes 1 ft on each edge, as shown.

a. Cubes in the bottom layer _____

b. Number of layers _____

c. Volume of the chest ____ ft^3

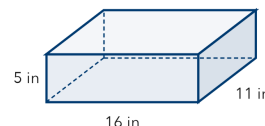


- 3.
- Fish Tank.**
- A fish tank is 16 in long, 11 in wide, and 5 in tall.

Hint Find the base first. The height counts the layers.

a. Area of the base ____ in^2

b. Volume of the tank ____ in^3



Show your work.

- 4.
- Locker Space.**
- A locker is 12 in wide, 15 in deep, and 48 in tall. Find its volume.

Hint Base = width $\times$ depth. Then multiply by the height.

Show your work.

Answer _____ in^3

- 5.
- Sandbox Fill.**
- A sandbox is 26 ft long and 11 ft wide. Sand is poured in 2 ft deep. How much sand is needed?

Hint The depth of the sand is the height.

Show your work.

Answer _____ ft^3

6. **Ian's Box.** A box is 17 cm long, 13 cm wide, and 5 cm tall. Ian says its volume is 221 cm^3 . Is Ian right?

Hint How many edges did Ian use? A prism has three.

Show your work.

True volume _____

7. **Two Cartons.** Carton A is 18 cm by 5 cm by 11 cm. Carton B is 15 cm by 11 cm by 5 cm. Which carton holds more? How much more?

Show your work.

Carton, how much _____

8. **Aquarium.** An aquarium is 60 cm long, 30 cm wide, and 40 cm tall. One liter of water fills $1,000\text{ cm}^3$.
- a. Find the volume. _____ cm^3
- b. How many liters fill it? _____ L
- c. It is filled only 35 cm deep. How many liters? _____ L

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Tomás's Work.** A crate is 15 ft long, 11 ft wide, and 4 ft tall. Tomás wrote: " $V = 15 + 11 + 4 = 30\text{ ft}^3$." Explain the mistake and find the correct volume.

TEST PRACTICE

choose the best answer

10. A prism is 3 cm by 4 cm by 5 cm. Its volume is –
- A** 12 cm^3 **B** 47 cm^3 **C** 60 cm^3 **D** 60 cm^2
11. Volume is measured in –
- A** units **B** square units **C** cubic units **D** degrees

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

60 C	572 A	8,640 P	221 Q	1,105 C	165 I	63 Y	880 A	72 T	286 Z
#2c	#3b	#4	#5	#6	#7	#8b	#8c		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____